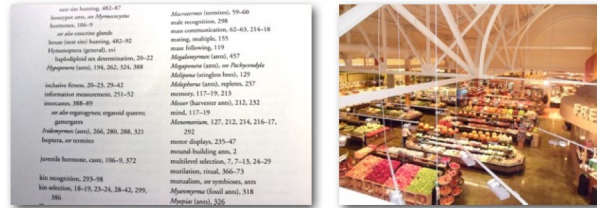


Data structures

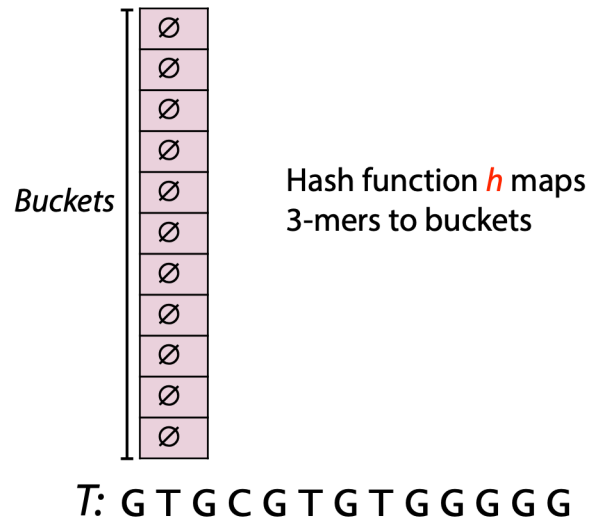


Multimap

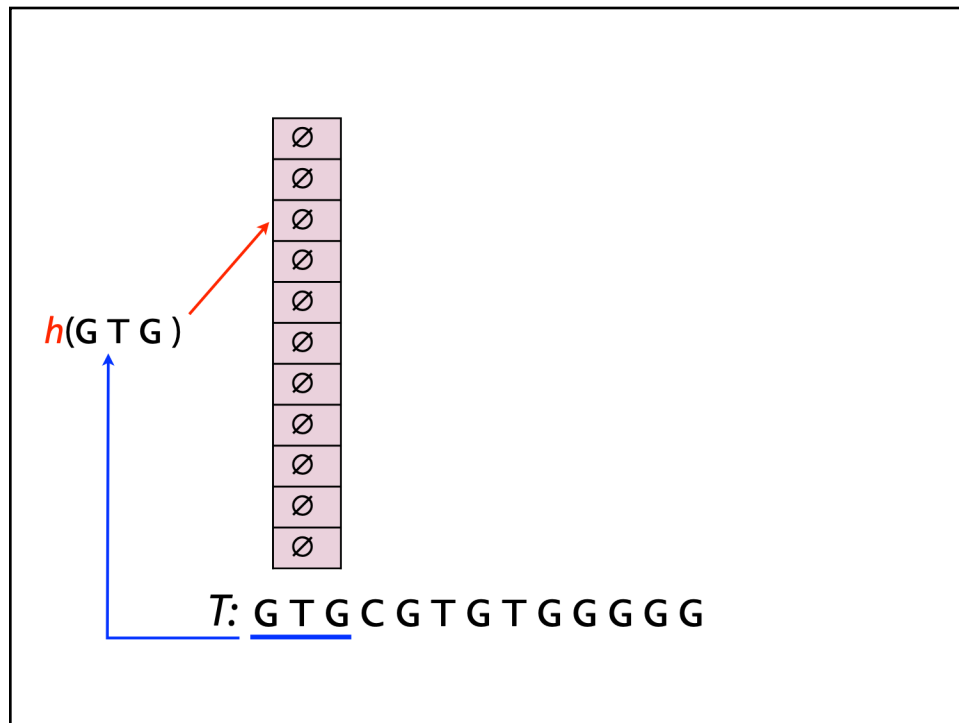
T: C G T G C G T G C T T

208

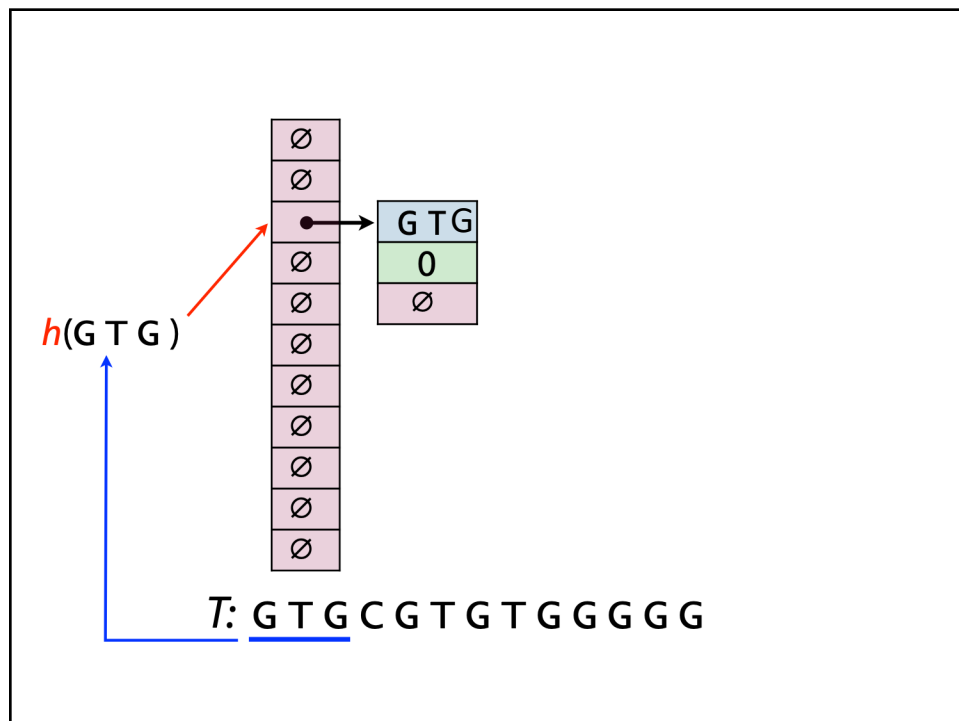
Hash table



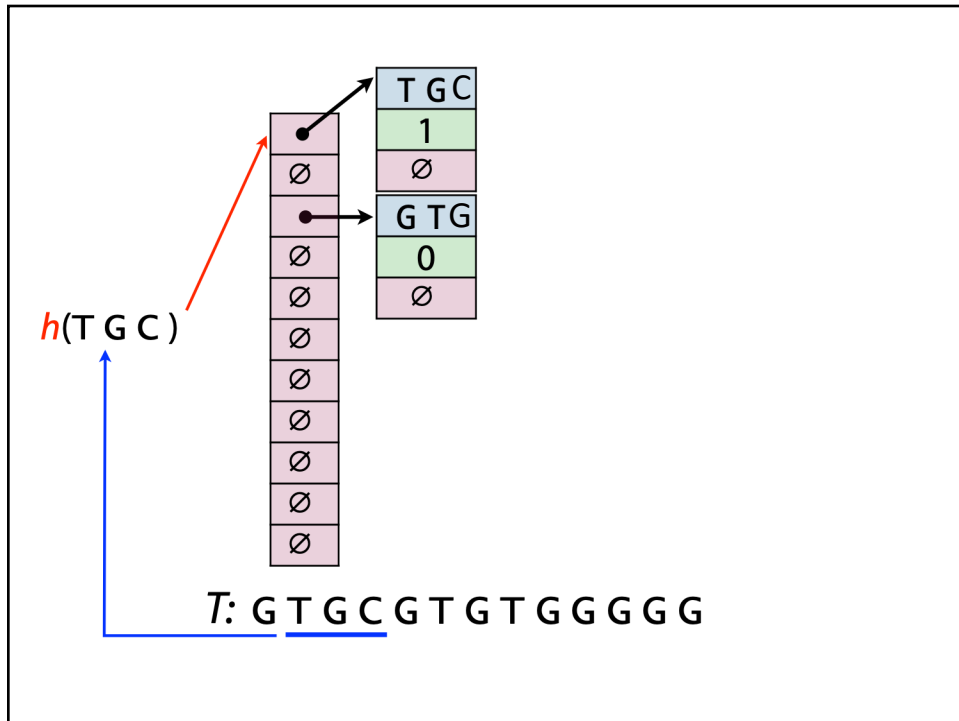
209



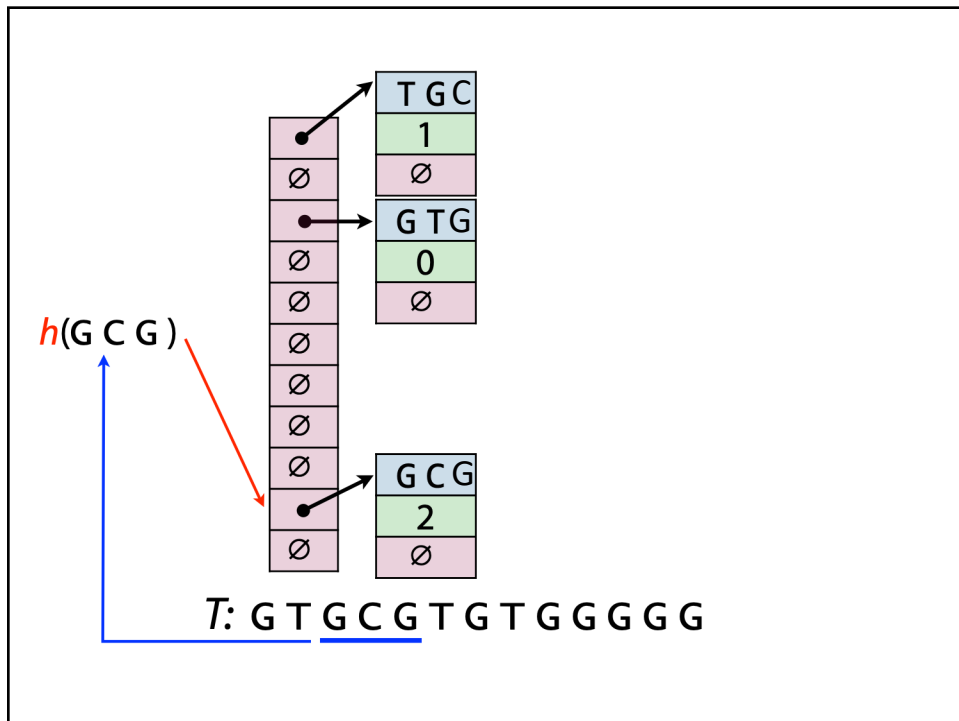
210



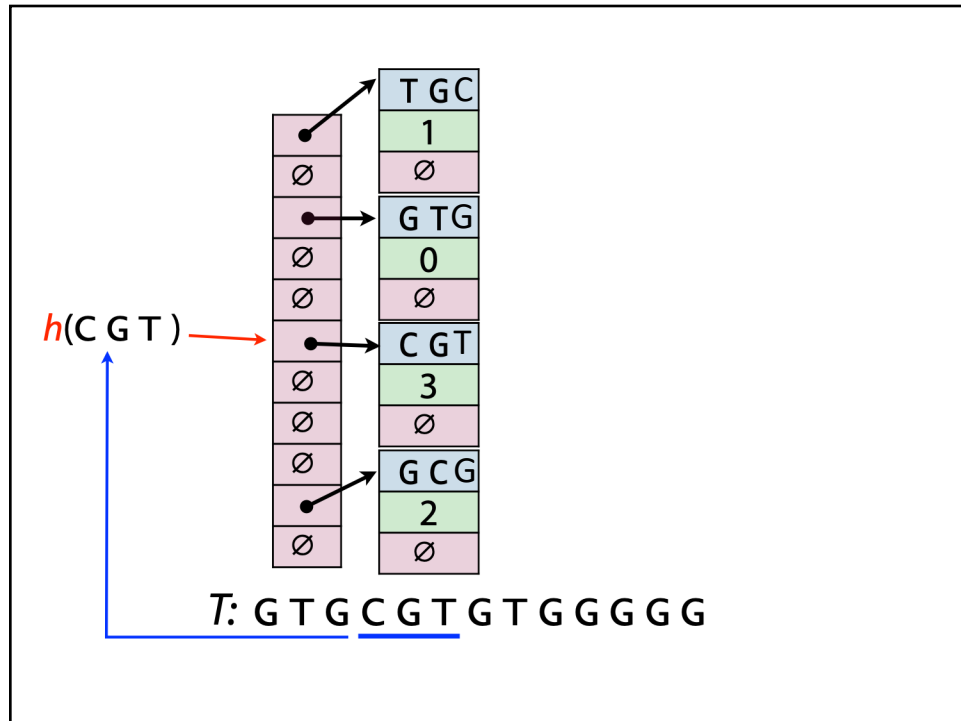
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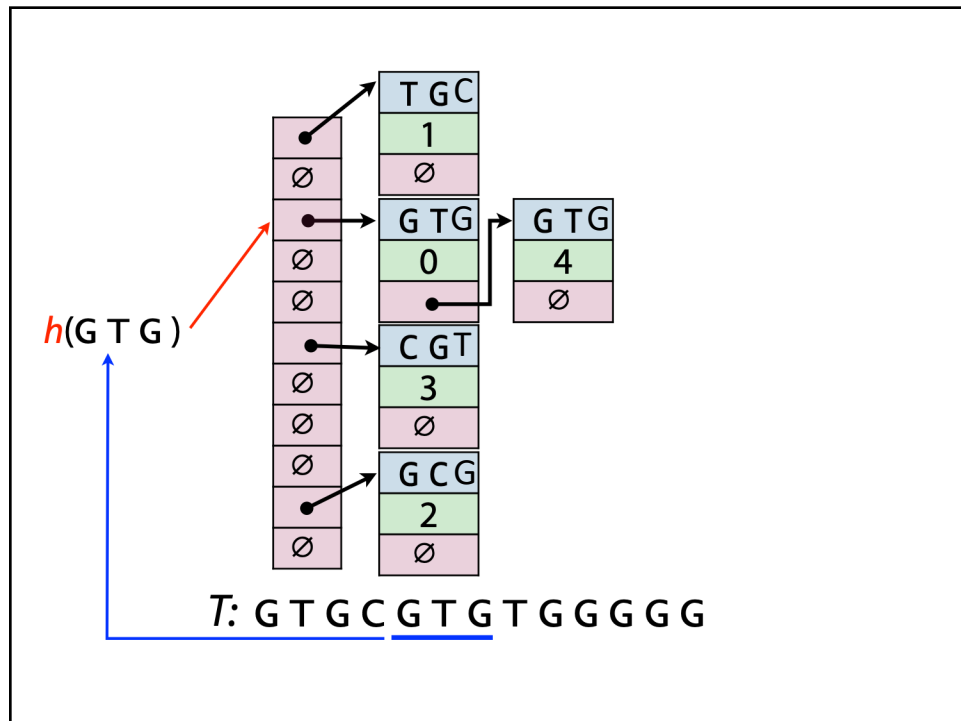
212



213



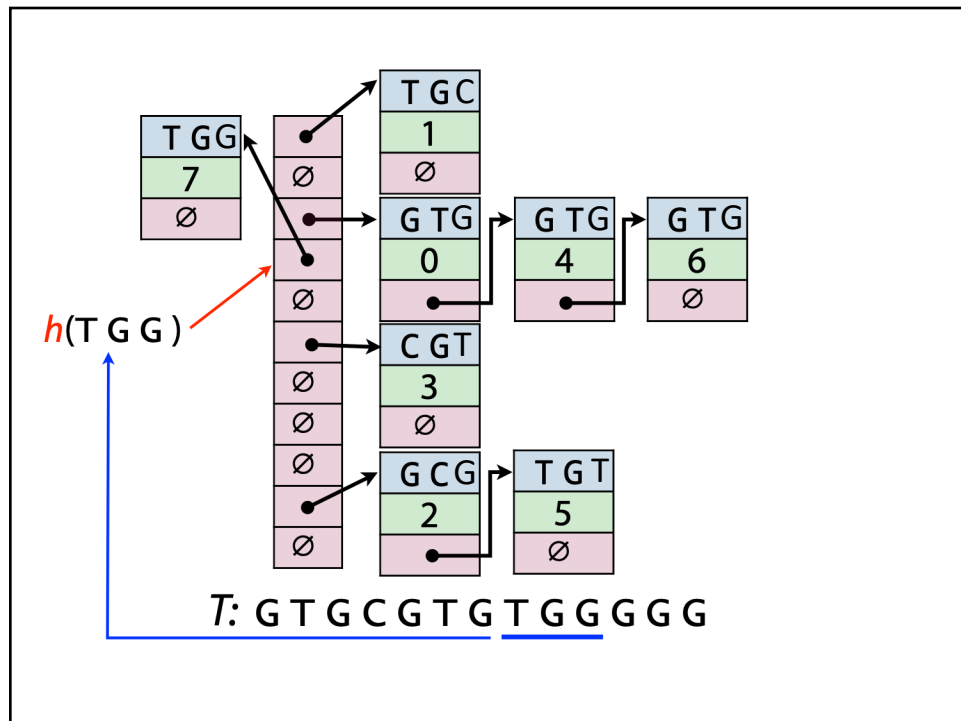
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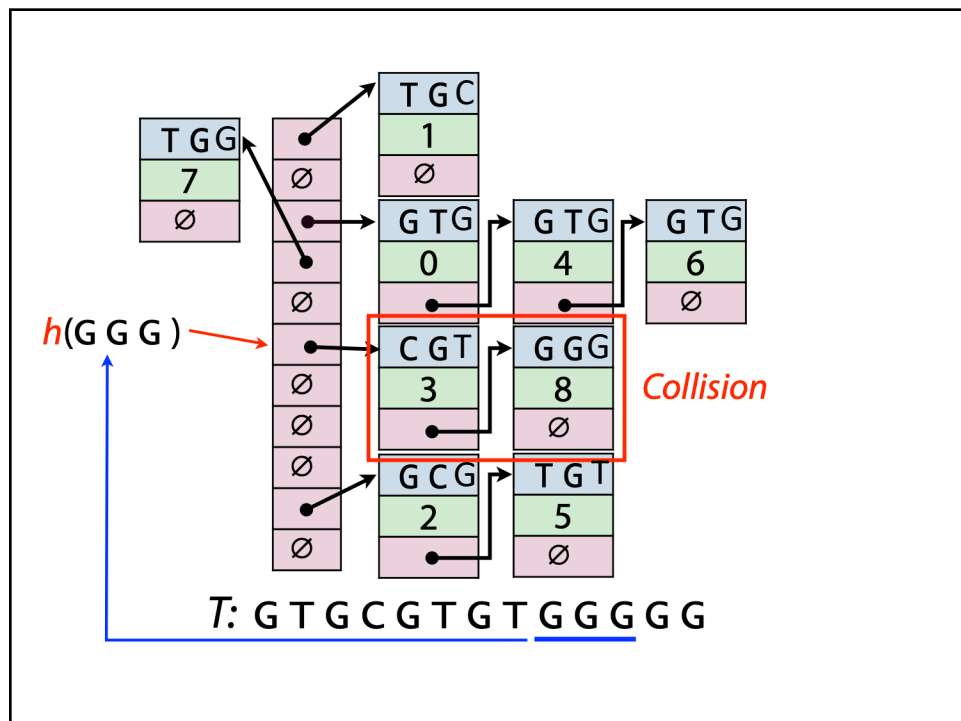
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216

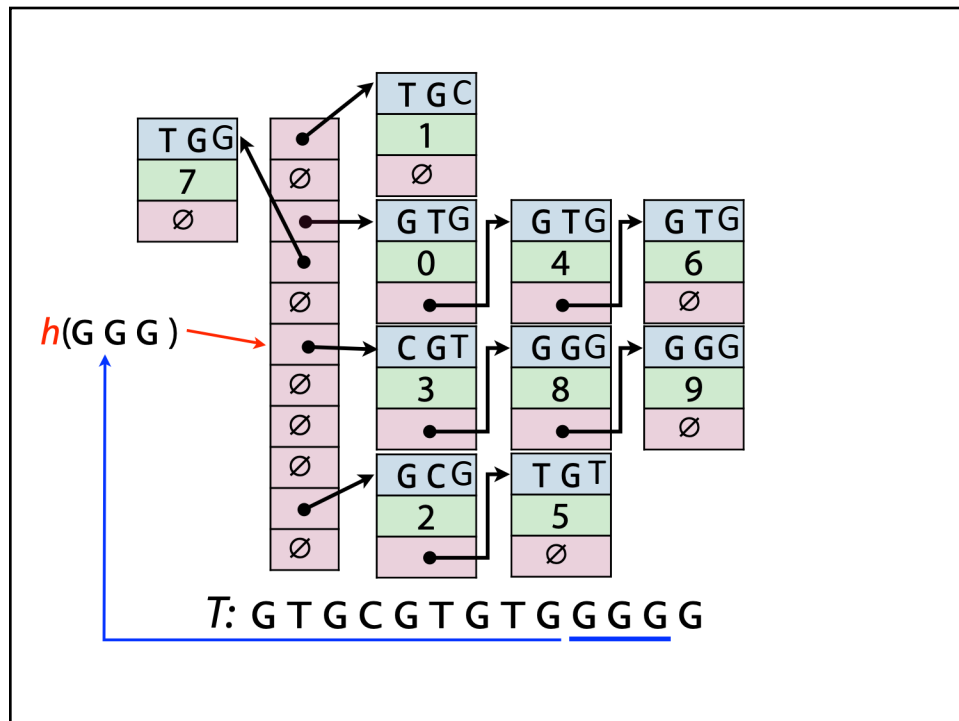
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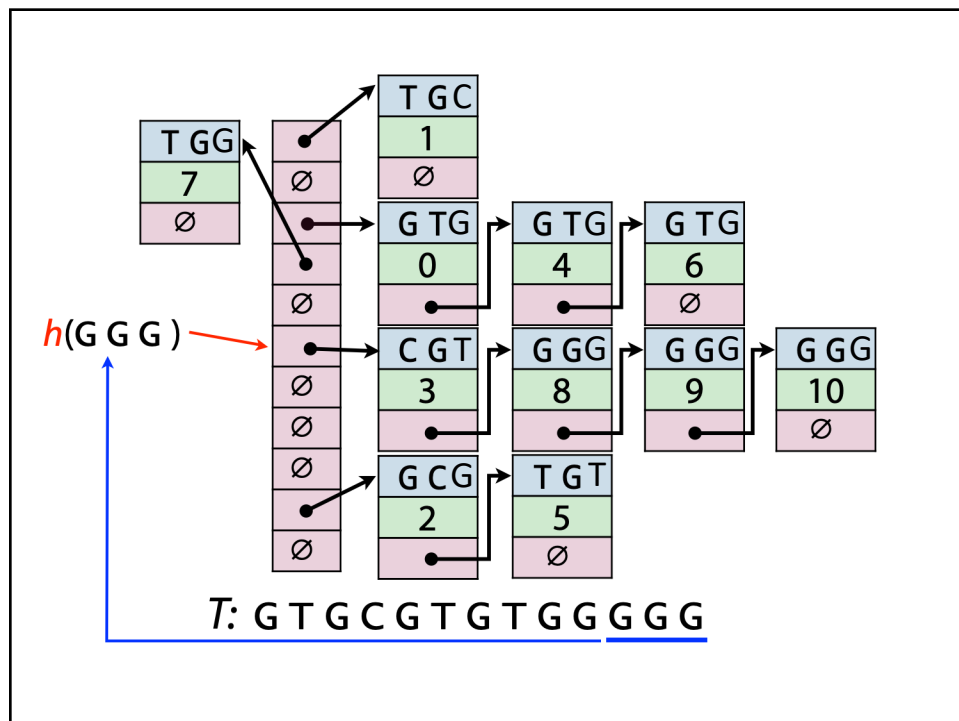
218



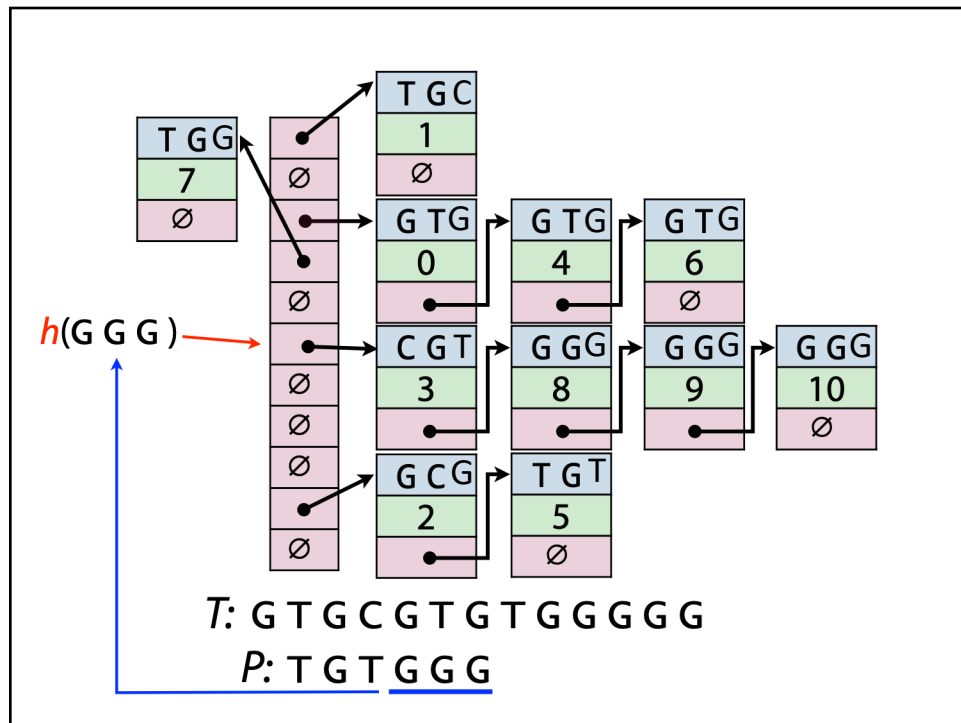
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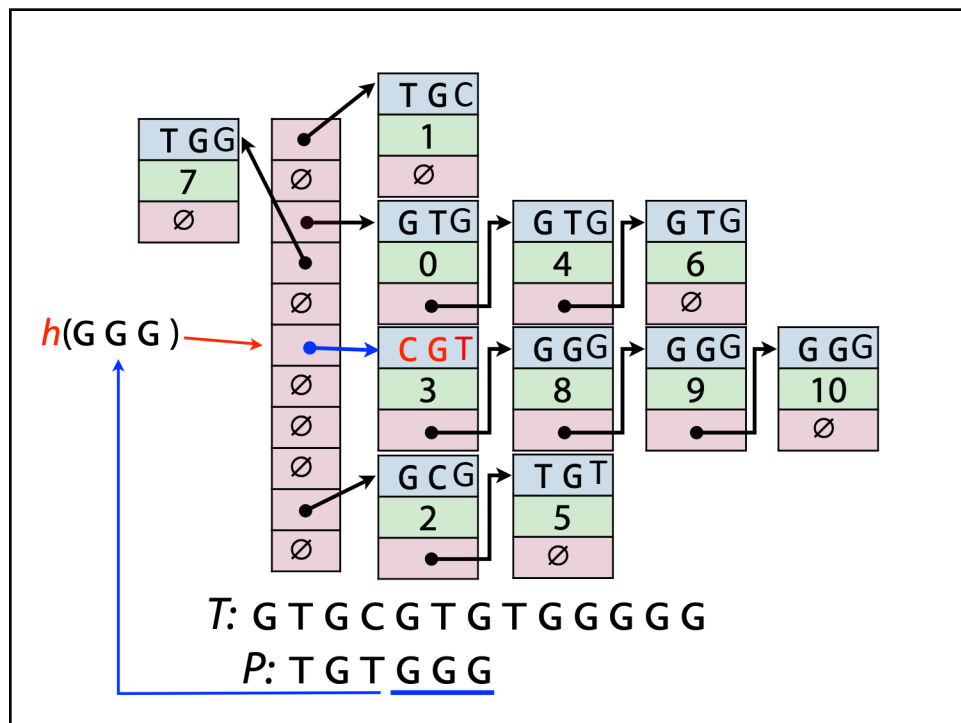
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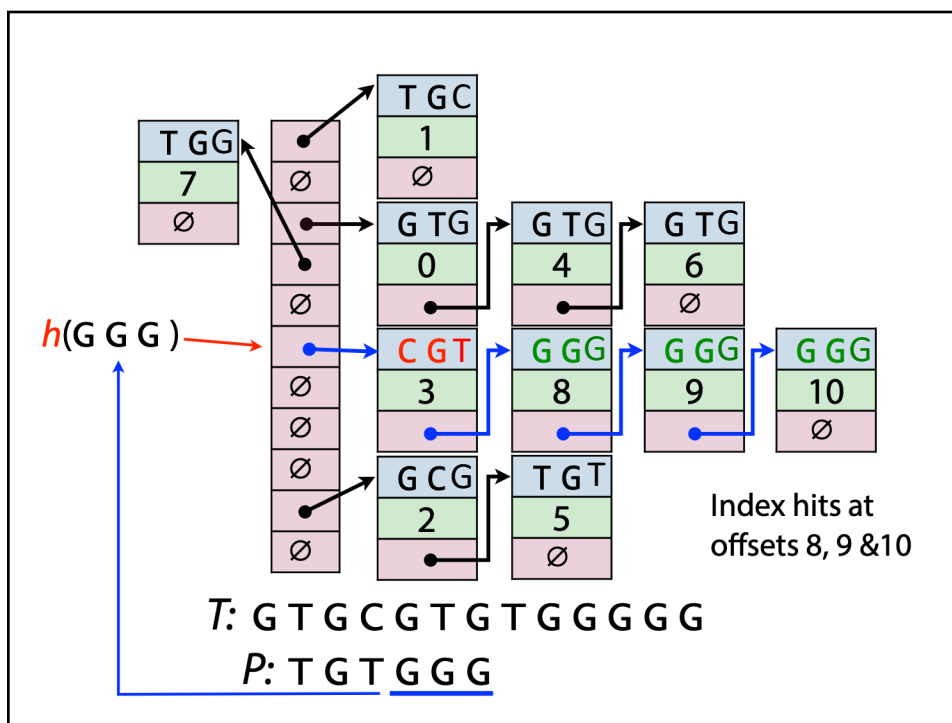
221



222

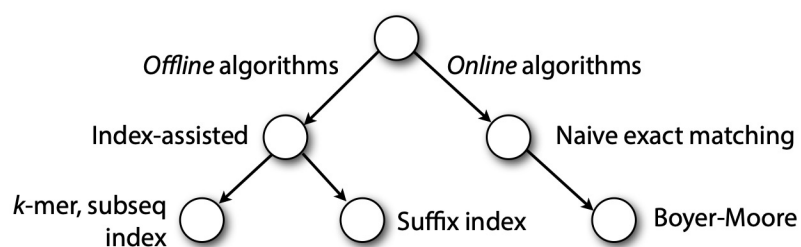


223



226

Exact Matching



227

Approximate matching

Read

CTCAAACTCTGACCTTTGGTGATCCACCCGCTAGGCCTTC

Reference

GATCACAGGCTATCACCTATTAACCACTACGGAGCTCTCATGCTTTGGTATTTT
CGTCTGGGGGTATGACGGGATAGCATTGCGAGACGCTGGAGCCGAGCACCTATGTC
GCAGTATCTGCTTTGATTCTGCCTCATCTATTATTTATCGACCTACGTTCAATATT
ACAGCGAACATACCTACTAAAGTGTGTTAATTAAATGCTTTGAGGACATAATAATA
ACAATTGAATGCTGACAGGCACCTTCCACAGACATCATACAAAAATTTCCACCA
AACCCCTCCCTCCCGCTTCTGGCCACAGCCTTCTGCGAAACCCCAAAA
ACAAAGAACCTAACACAGCCTAACCTTTTCAATTTTCTGCGGTATGCAC
TTTTACAGTCACCCCACTAACCTTTATTTTCCCTCCCTGCTCATACTACTAAT
CTCATCAATACAAACCCCGCCATCTACCCAGACACACAGCCTTAAACCCATA
CCCGAACCAACCAACCCCAAACTACCCCGCACAGTTTATGATGCTCTCTCAAA
GCAATACACTGACCGCTCAAACTCTGGATTTGGATCCACAGCCTTGGCTAAA
CTAGCCTTTCTATTAGCTCTTAGAAGATTACATGCAAGCATCCCTCCAGTGA
TCACCTCTAAATCACACGATCTAAGGAACAGCATCAAGCAGCCTAATGACATC
AAACGCTTAGCTAGCCACACCTACCGGAAACAGCAGTATTAACTTAGCAATAA
ACGAAAGTTTAACTAGCTATACCTACCGAGGTGGTCAATTTCTACAGCCACCGC
GGTCACAGATTAAACCAAGTCAATCTACCGGGTAAGAGTGTATAGATCACCCCT
TCCCAATAAAGCTAAACTCAGCTGCTGTAAAAAAGCTCCAGCTTACAAATAGAC
TACGAAAGTGGCTTTAACAATCTGAACCTCAATAGCTAAGCTTGGGATTAGA
TACCCCACTATGCTTAGCCCTAACTCAACCTTAACTCAACCTTGGCCAGAA
CACTACGAGCCACAGCTTAAACTCAAGGACCTGGCGGTGCTTATCTAGAGG
AGCCTGTTCTGTAATCGATAAACCCGATCAACCTCACCACCTCTGCTATATA
CCGCCATCTCAGCAACCTGATGAAGGCTACAAAGTAAGCGAAGTACCTGAG
ACGTTAGGTCAAGGTGAGCCATGAGGTGGCAAGAAATGGGCTACATTTCTGAG
AAACTACGATAGCCTTATGAACCTTAAAGGTGCAAGGTGGATTAGCAGTAAT
AGTAGAGTCTTAGTTGAACAGGGCCCTGAAGCGGTACACACCGCCGTCACCT
AAGTATACCTCAAGGACATTTAACTAAACCCCTACGCTTATATAGAGGAGACAA
CGTAACCTCAACTCTGCTTTGGTGATCCACCGCTTGGCTACTGCTAATGAAG
AAGCAACCACTTACCTTAGGAGATTTCACTTAACTTGGCGCTGAGCTAACTTA
GCCCAACCACTCCACCTTACTACCAAGCAACCTTAGCCTAACCACTTATCCCAATAA
AGTATAGGCGATAGAAATTGAACCTGGCGCAATAGATATAGTACCGCAAGGAAAGATG
AAAAATTATAACCAAGCTAATATAGCAAGGACTAACCTTATACCTCTGCTCATATGAA
TTAACTAGAAATACTTTGCAAGGAGAGCCAAAGCTAAGACCCCGAAACCAAGAGCT
ACCTAGAACAGCTAAAGAGCACACCGCTATGTAGCAAAATAGTGGGAAGATTATA
GGTAGAGGCGCAAACTACCGAGCCGCTGATAGCTGGTGTCCCAAGATAGATCTTAG
TTCAACTTTAAATTTGCCCAAGAACCTCTAAATCCCTTGTAAATTTAACTGTTAGTC

Differences between read and reference occur because of...

1. Sequencing error
2. Natural variation

228

Approximate matching

T: GGAAAAAGAGGTACGGCGTTTAAACAGTAG

P: GTAAACGGCG

↑
Mismatch
(Substitution)

229

Approximate matching

T: GGAAAAGAGGTAGC-GCGTTTAACAGTAG
 | | | | | | | |
P: GTAGC-**G**GCG
 ↑
 Insertion

230

Approximate matching

T: GGAAAAGAGGT**A**GCGGCGTTTAACAGTAG
 | | | | | | | |
P: GT-**G**CGGCG
 ↑
 Deletion

231

Hamming distance

For X & Y where $|X| = |Y|$, *hamming distance* =
minimum # substitutions needed to turn one into the other

X: G A G G T A G C G G C G T T T A A C
 | | | | | | | | | | | | | | |
 Y: G T G G T A A C G G G G T T T A A C *Hamming distance = 3*

232

Edit distance

(AKA Levenshtein distance)

For X & Y , *edit distance* = minimum # edits (substitutions, insertions, deletions) needed to turn one into the other

X: T G G C C G C G C A A A A C A G C
 | | | | | | | | | | | | | | |
 Y: T G A C C G C G C A A A A - C A G C *Edit distance = 2*

X: G C G T A T G C G G C T A - A C G C
 | | | | | | | | | | | | | | |
 Y: G C - T A T G C G G C T A T A C G C *Edit distance = 2*

233

Approximate matching

Wanted: way to apply exact matching algorithms to approximate matching problems

234

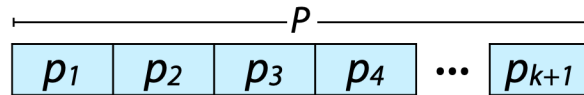
Approximate matching



If P occurs in T with 1 edit, then u or v appears with no edits

235

Approximate matching



If P occurs in T with up to k edits, then at least one of $p_1, p_2, \dots, p_{k+1}$ must appear with 0 edits

236

Approximate matching



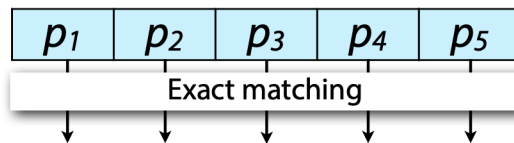
237

Pigeonhole principle



238

Pigeonhole principle


 T

239

Pigeonhole principle

p_4
match

T

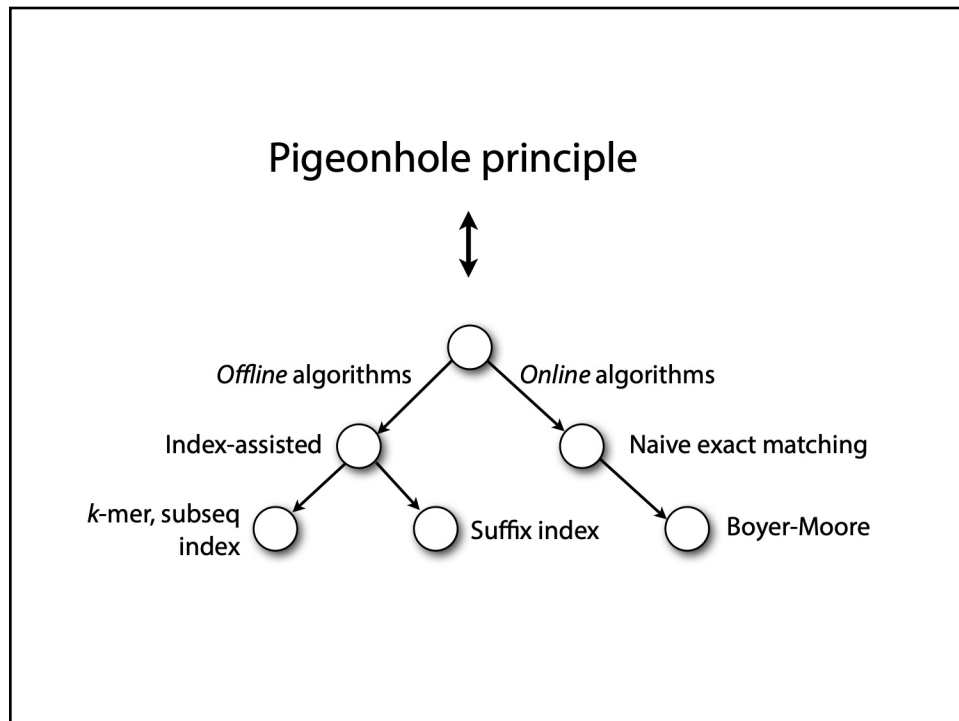
240

Pigeonhole principle

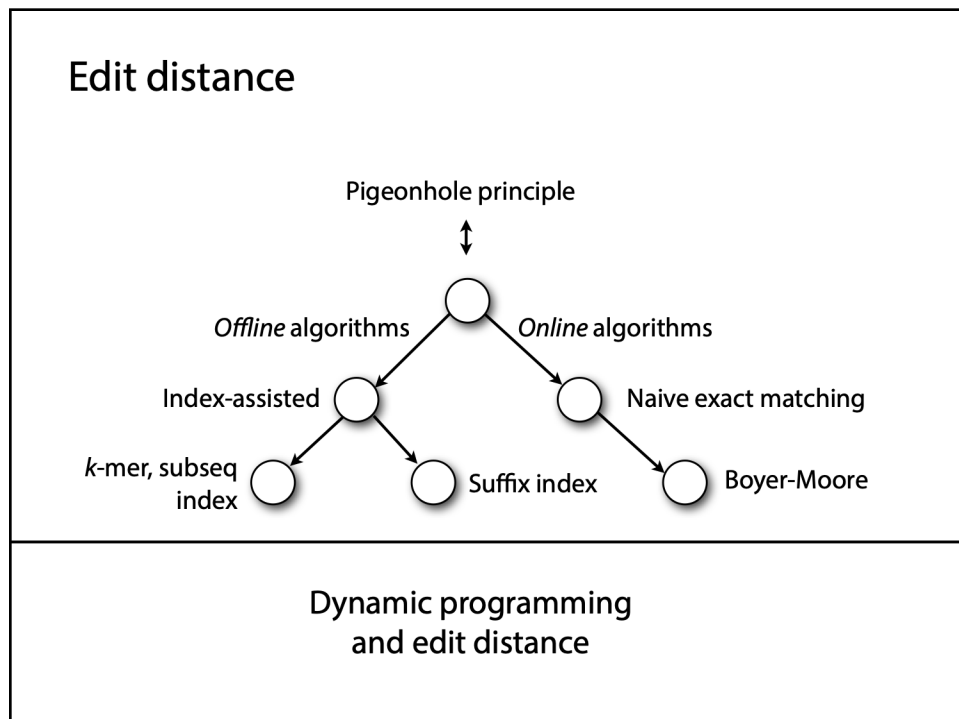
p_1 p_2 p_3 p_4 p_5
↑ ↑ ↑ ↓ ↑
verify verify verify match verify

T

241



242



243

Edit distance

For X, Y where $|X| = |Y|$, *hamming distance* = minimum # substitutions needed to turn one into the other

For X, Y , *edit distance* = minimum # edits (substitutions, insertions, deletions) needed to turn one into the other

244

Edit distance

If $|X| = |Y|$ what can we say about the relationship between **editDistance**(X, Y) and **hammingDistance**(X, Y)?

$$\text{editDistance}(X, Y) \leq \text{hammingDistance}(X, Y)$$

```

X: G C G T A T G C G G C T A - A C G C
   | |   | | | | | | | | | | | |
Y: G C - T A T G C G G C T A T A C G C

```

245

Edit distance

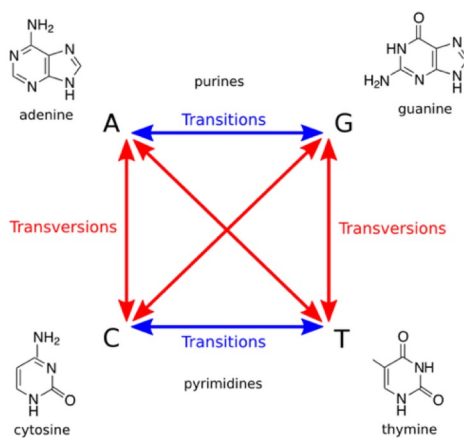
If x and y are different lengths, what can we say about $\text{editDistance}(X, Y)$?

$$\text{editDistance}(X, Y) \geq ||X| - |Y||$$

$X: ? ?$

$Y: ? ? ? ?$

246



Human *transition to transversion ratio* (AKA *ti/tv*) is ~2.1

247

```

GGGTA G CGGGTTTAAC
||| ||| ||| ||| |||
GGGTA A CGGGTTTAAC

```

Human substitution rate ≈ 1 in 1,000

```

GGGTA G CGGGTTTAAC
||| ||| ||| ||| |||
GGGTA - - GGGTTTAAC

```

Small-gap rate is ≈ 1 in 3,000

248

Penalty matrix

	A	C	G	T	-
A	0	4	2	4	8
C	4	0	4	2	8
G	2	4	0	4	8
T	4	2	4	0	8
-	8	8	8	8	

2 Transitions (A ↔ G, C ↔ T)

4 Transversions

8 Gaps

249

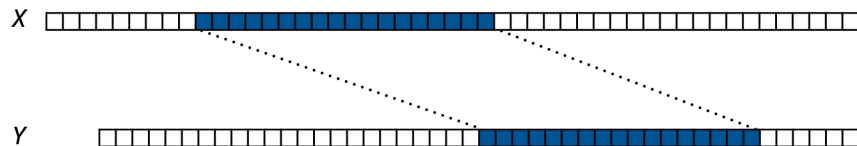
Scoring matrix: matches are positive, differences negative

	A	C	G	T	-
A	2	-4	-4	-4	-6
C	-4	2	-4	-4	-6
G	-4	-4	2	-4	-6
T	-4	-4	-4	2	-6
-	-6	-6	-6	-6	

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Local alignment

Find the most similar *pair of substrings* from X and Y



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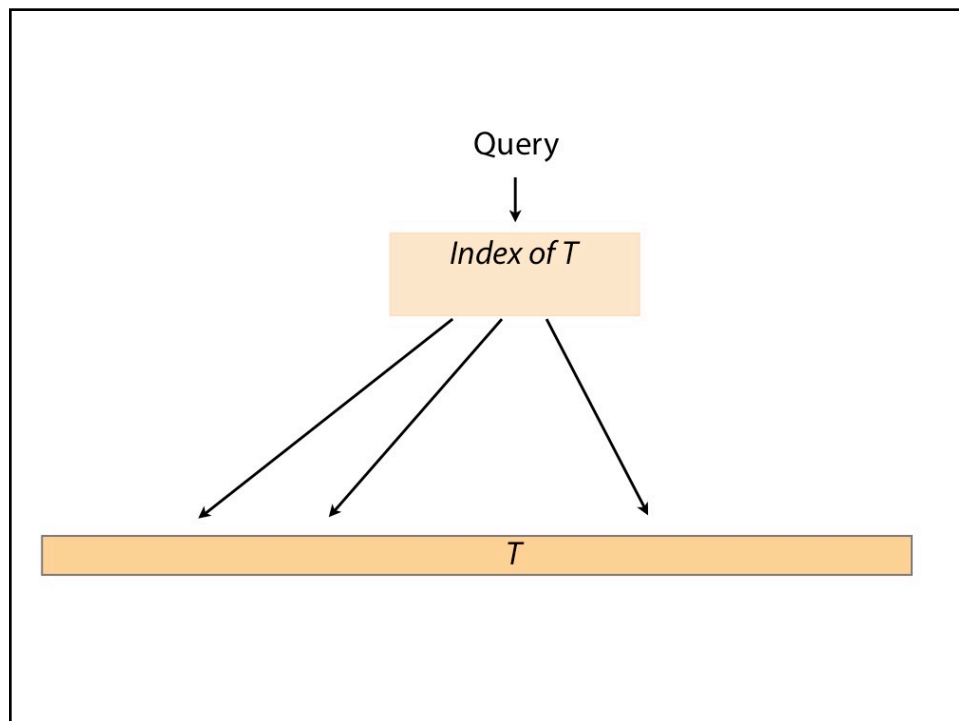
Local alignment

Find the most similar *pair of substrings* from X and Y

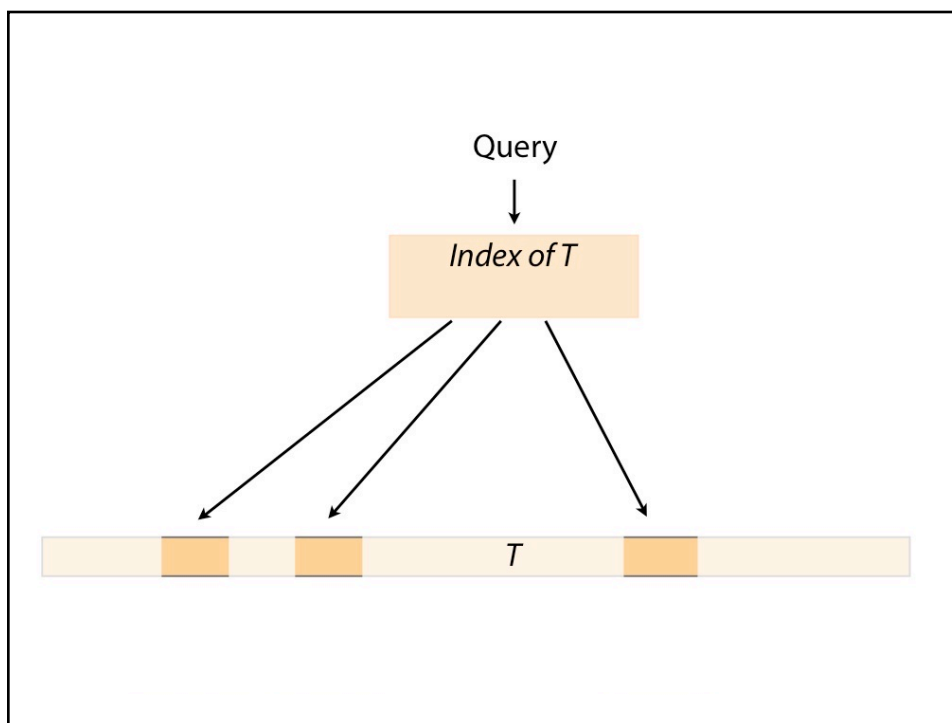
X he_will_after_his_sour_fashion_tell_you

Y struts_and_frets_his_hour_upon_the_stage

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J. Mol. Biol. (1990) 215, 403–410

Basic Local Alignment Search Tool

Stephen F. Altschul¹, Warren Gish¹, Webb Miller²
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¹National Center for Biotechnology Information
National Library of Medicine, National Institutes of Health
Bethesda, MD 20894, U.S.A.

²Department of Computer Science
The Pennsylvania State University, University Park, PA 16802, U.S.A.

³Department of Computer Science
University of Arizona, Tucson, AZ 85721, U.S.A.

(Received 26 February 1990; accepted 15 May 1990)

A new approach to rapid sequence comparison, basic local alignment search tool (BLAST), directly approximates alignments that optimize a measure of local similarity, the maximal segment pair (MSP) score. Recent mathematical results on the stochastic properties of MSP scores allow an analysis of the performance of this method as well as the statistical significance of alignments it generates. The basic algorithm is simple and robust; it can be implemented in a number of ways and applied in a variety of contexts including straight-forward DNA and protein sequence database searches, motif searches, gene identification searches, and in the analysis of multiple regions of similarity in long DNA sequences. In addition to its flexibility and tractability to mathematical analysis, BLAST is an order of magnitude faster than existing sequence comparison tools of comparable sensitivity.

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NIH U.S. National Library of Medicine NCBI National Center for Biotechnology Information

BLAST® [Home](#)

BLAST finds regions of similarity between biological sequences. [more...](#)

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BLAST Assembled Genomes

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Enter organism name or id—completions will be suggested **GO**

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- [Chicken](#)
- [Zebrafish](#)
- [Clawed frog](#)
- [Arabidopsis](#)
- [Rice](#)
- [Yeast](#)
- [Microbes](#)

Basic BLAST

Choose a BLAST program to run.

- nucleotide blast** Search a **nucleotide** database using a **nucleotide** query
Algorithms: blastn, megablast, discontinuous megablast
- protein blast** Search **protein** database using a **protein** query
Algorithms: blastp, psi-blast, phi-blast, delta-blast
- blastx** Search **protein** database using a **translated nucleotide** query
- tblastn** Search **translated nucleotide** database using a **protein** query
- tblastx** Search **translated nucleotide** database using a **translated nucleotide** query

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Pre-computed BLAST databases

- separate sequence identifiers from sequences
- break sequences into "words" (like kmers)
- you can also format your own database (later)

File Name	Content Description
16SMicrobial.tar.gz	Bacterial and Archaeal 16S rRNA sequences from BioProjects 33175 and 33117
FASTA/	Subdirectory for FASTA formatted sequences
README	README for this subdirectory (this file)
Representative_Genomes.*tar.gz	Representative bacterial/archaeal genomes database
cdd_delta.tar.gz	Conserved Domain Database sequences for use with stand alone deltablast
cloud/	Subdirectory of databases for BLAST AMI; see http://1.usa.gov/TJAnEt
env_nr.*tar.gz	Protein sequences for metagenomes
env_nt.*tar.gz	Nucleotide sequences for metagenomes
est.tar.gz	This file requires est_human.*tar.gz, est_mouse.*tar.gz, and est_others.*tar.gz files to function. It contains the est.nal alias so that searches against est (-db est) will include est_human, est_mouse and est_others.
est_human.*tar.gz	Human subset of the est database from the est division of GenBank, EMBL and DDBJ.
est_mouse.*tar.gz	Mouse subset of the est database
est_others.*tar.gz	Non-human and non-mouse subset of the est database
gss.*tar.gz	Sequences from the GSS division of GenBank, EMBL, and DDBJ
htgs.*tar.gz	Sequences from the HTG division of GenBank, EMBL, and DDBJ
human_genomic.*tar.gz	Human RefSeq (NC_####) chromosome records with gap-adjusted concatenated NT_contigs
nr.*tar.gz	Non-redundant protein sequences from GenPept, Swissprot, PIR, PDF, PDB, and NCBI RefSeq
nt.*tar.gz	Partially non-redundant nucleotide sequences from all traditional divisions of GenBank, EMBL, and DDBJ excluding GSS, STS, PAT, EST, HTG, and WGS
other_genomic.*tar.gz	RefSeq chromosome records (NC_####) for non-human organisms
pataa.*tar.gz	Patent protein sequences
patnt.*tar.gz	Patent nucleotide sequences. Both patent databases are directly from the USPTO, or from the EPO/JPO via EMBL/DDBJ
pdtaa.*tar.gz	Sequences for the protein structure from the Protein Data Bank
pdnt.*tar.gz	Sequences for the nucleotide structure from the Protein Data Bank. They are NOT the protein coding sequences for the corresponding pdtaa entries.
refseq_genomic.*tar.gz	NCBI genomic reference sequences
refseq_protein.*tar.gz	NCBI protein reference sequences
refseq_rna.*tar.gz	NCBI Transcript reference sequences
sts.*tar.gz	Sequences from the STS division of GenBank, EMBL, and DDBJ
swissprot.tar.gz	Swiss-Prot sequence database (last major update)
taxdb.tar.gz	Additional taxonomy information for the databases listed here
tsa_nt.*tar.gz	providing common and scientific names
vector.tar.gz	Sequences from the TSA division of GenBank, EMBL, and DDBJ
wgs.*tar.gz	Vector sequences from 2010, see Note 2 in section 4.
	Sequences from Whole Genome Shotgun assemblies

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How does BLAST 1990 work?

1. Query sequence is also broken into “words” of length w ($w=11$ for DNA and 3 for protein)
2. Align all words from the query to those of database and keep only ones that produce an alignment score above the threshold “ T ” (scores are calculated using a substitution matrix).
3. Extend each of the hits in both directions, and identify the maximum segment pairs “ MSP ” (also called High Scoring Pairs, the pair of identical length segments chosen from 2 sequences that score above a cutoff score “ S ”)

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BLAST E-values

E-value = the number of HSPs having alignment score S (or higher) expected to occur **by chance**.

- Smaller E-value, more significant in statistics
- Bigger E-value, less significant
- Over 1 means expect this totally by chance (not significant at all!)

The expected number of HSPs with the score at least S is :

$$E = K * n * m * e^{-\lambda S}$$

K, λ are constant depending on model

n, m are the length of query and sequence

E-values quickly drop off for better alignment bits scores

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Example

Alignment starts with initial word of 11

```
ACACTGAGTGA
|||||
ACACTGAGTGA
```

Extension to the left has no mismatches, no penalty points
Extension to the right has mismatches and penalty points

```
GCACCTTTGCCACACTGAGTGAAGTGTCTATG
|||||
GCACCTTTGCCACACTGAGTGAAGTGTCTATG
```

Extension to the left has no penalty points and can continue to grow
Extension to the right accumulates too many mismatch penalty points; extension in this direction stops

```
CAACCTCAAGGGACCTTTGCCACACTGAGTGAAGTGTCTATGCTCCTTTGGGG
|||||
CAACCTCAAGGGACCTTTGCCACACTGAGTGAAGTGTCTATGCTCCTTTGGGG
```

If left side cannot grow any more, the final alignment looks like this:

```
CAACCTCAAGGGACCTTTGCCACACTGAGTGAAGTGTCTATG
|||||
CAACCTCAAGGGACCTTTGCCACACTGAGTGAAGTGTCTATG
```

Figure 3.1 Simple extension example for BLASTN. Starting with an initial match of “words,” BLAST extends the alignment between query and hit, keeping track of penalty points against, and increasing significance for, extending the alignment.

practical_bioinformatics_chapter3

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New in BLAST2 (1997)

Gapped BLAST and PSI-BLAST: a new generation of protein database search programs

Stephen F. Altschul^{*}, Thomas L. Madden, Alejandro A. Schäffer¹, Jinghui Zhang, Zheng Zhang², Webb Miller² and David J. Lipman

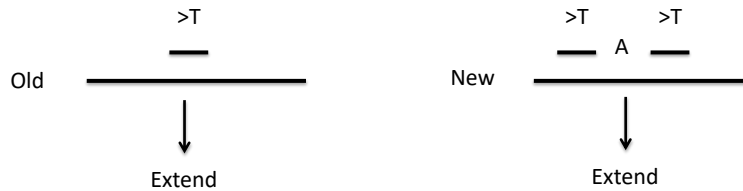
National Center for Biotechnology Information, National Library of Medicine, National Institutes of Health, Bethesda, MD 20894, USA, ¹Laboratory of Genetic Disease Research, National Human Genome Research Institute, National Institutes of Health, Bethesda, MD 20892, USA and ²Department of Computer Science and Engineering, Pennsylvania State University, University Park, PA 16802, USA

Received June 20, 1997; Revised and Accepted July 16, 1997

- Old BLAST: seeks short word pairs whose aligned score is at least T. Each such ‘hit’ is then extended, to test whether it is contained within a high-scoring alignment.
- New BLAST: requires the existence of two non-overlapping word pairs on the same diagonal, and within a distance A of one another, before an extension is invoked. T is lower to achieve the same sensitivity, but the number of actual extensions is much lower, and hence, **faster**.

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New in BLAST2 (1997)



- Old BLAST: seeks short word pairs whose aligned score is at least T . Each such 'hit' is then extended, to test whether it is contained within a high-scoring alignment.
- New BLAST: requires the existence of two non-overlapping word pairs on the same diagonal, and within a distance A of one another, before an extension is invoked. T is lower to achieve the same sensitivity, but the number of actual extensions is much lower, and hence, **faster**.

Alstchul, 1997

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• 2. Gapped alignments:

Old BLAST: treats gapped alignments implicitly by locating, in many cases, several distinct HSPs involving the same database sequence, and calculating a statistical assessment of the combined result. So more than one is needed to pass the threshold.

New BLAST: triggers a gapped extension for any HSP that exceeds a moderate score S_g . By seeking a single gapped alignment, rather than a collection of ungapped ones, only one of the constituent HSPs needs to be located for the combined result to be generated successfully – so we can use higher T for the hit stage. Also, considers only alignments that drop in score no more than X_g below the best score yet seen.

Alstchul, 1997

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In summary, the new gapped BLAST algorithm requires two non-overlapping hits of score at least T , within a distance A of one another, to invoke an ungapped extension of the second hit. If the HSP generated has normalized score at least S_g bits, then a gapped extension is triggered. The resulting gapped alignment is reported only if it has an E-value low enough to be of interest.

Alstchul, 1997

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Substitution Matrix DNA

	A	C	G	T
A	5	-4	-4	-4
C	-4	5	-4	-4
G	-4	-4	5	-4
T	-4	-4	-4	5

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Quite Similar Sequences

Query: HBA_HUMAN Hemoglobin alpha subunit
Sbjct: MYG_HUMAN Myoglobin

Score = 51.2 bits (121), Expect = 1e-07,
Identities = 38/146 (26%), Positives = 58/146 (39%), Gaps = 6/146 (4%)

```

Query  2  LSPADKTNVKAANGKVGAHAGEYGAEALERMFLSFPTTKTYFPHF-----DLSHGSAQV  55
      LS  +  V  WGKV A  +G E L R+F  P T  F  F      D  S  +
Sbjct  3  LSDGEWQLVLNVWGKVEADIPGHGQEVLRIRLFKGHPEFLEKFDKFKHLKSEDEMKASEDL  62

Query  56  KGHGKKVADALTNVAHVDDMPNALSALSDDLHAKLRVDPVNFKLLSHCLLVTLAAHLPA  115
      K HG  V  AL  +      + L+  HA K ++      + +S C++  L  +  P
Sbjct  63  KKHGATVLTALGGILKKKGHHEAEIKPLAQSHATKHKIPVKYLEFISECIIQVLQSKHPG  122

Query  116  EFTPAVHASLDKFLASVSTVLTSKYR  141
      +F      +++K L      + S Y+
Sbjct  123  DFGADAQGAMNKALELFRKDMASNYK  148

```

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Not similar sequences

Query: HBA_HUMAN Hemoglobin alpha subunit
Sbjct: SPAC869.02c [Schizosaccharomyces pombe]

Score = 33.1 bits (74), Expect = 0.24
Identities = 27/95 (28%), Positives = 50/95 (52%), Gaps = 10/95 (10%)

```

Query  30  ERMFLSFPTTKTYFPHFDSLHSGSAQVKGHGKKVADALTNVAHVDDMPNALSALSDDLHAAH  89
      ++M ++P      P+F+ +H  +      + +A AL N  ++DD+  +LSA  D
Sbjct  59  QKMLGNYPEV---LPYFNKAHQISL--SQPRILAFALLNYAKNIDDL-TSLSAFMDQIVV  112

Query  90  K---LRVDPVNFKLLSHCLLVTLAAHLPAEF-TPA  120
      K  L++  ++ ++ HCLL T+  LP++  TPA
Sbjct  113  KHVGLQIKAEHYPIVGHCLLSTMQELLPSDVATPA  147

```

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